

ULTRON

Personal AI assistant for phone and laptop

Project documentation: features, architecture, structure, setup and testing

---

Python | SQLite | HTML / JavaScript | face-api.js | Google Gemini | Ollama / Qwen

Version: November 2026

Contents

1. Project overview **3**

1.1 Highlights . . . 3

1.2 Technology at a glance . . . 4

2. Features in detail **4**

2.1 Owner identity and security . . . 4

2.2 Interface: the six Infinity Stones . . . 5

2.3 Voice . . . 6

2.4 Commands ULTRON understands offline . . . 6

2.5 Artificial intelligence with Gemini . . . 7

2.6 Student tools . . . 7

2.7 Phone and laptop: the same features . . . 8

2.8 Sync between phone and laptop . . . 8

3. System architecture **9**

3.1 How the parts work together . . . 9

3.2 Life of a request . . . 10

3.3 Data model . . . 10

3.4 Server API . . . 10

4. Project structure **11**

5. ULTRON Core: the machine-learning layer **12**

5.1 The response contract . . . 12

5.2 System tools . . . 13

5.3 Persona and prompt framework . . . 13

5.4 Reliability and safety . . . 13

5.5 Engines . . . 14

5.6 Fine-tuning plan (optional) . . . 14

5.7 How quality is measured . . . 14

6. Setup and use **15**

6.1 Laptop (Windows) . . . 15

6.2 Phone (Android) . . . 15

6.3 Connecting the two devices . . . 15

6.4 Settings . . . 15

7. Testing and verification **16**

8. Limitations and future work **16**

8.1 Known limitations . . . 16

8.2 Possible next steps . . . 17

9. Glossary **17**

1. Project overview

ULTRON is a voice-controlled personal assistant built for a university student. One program runs on both an Android phone (through Termux) and a Windows laptop, shows the same futuristic interface on both, and keeps the data on the two devices in sync. It recognises its owner by face, voice or PIN, manages the college timetable, tasks, deadlines and study plan, reminds the owner at the right time, opens apps and websites, places phone calls, and talks back through an animated robot face.

Its intelligence works in two layers. Everyday commands are understood offline by ULTRON's own command parser, which is instant and free. Harder questions go to Google Gemini, but **only when the owner asks for it** by saying "use Gemini". Gemini can then think step by step, use ULTRON's tools, write and run code, search the web, look through the camera, and save new methods as reusable skills. A separate machine-learning core (ULTRON Core) defines a strict structured-output "brain" that can run on Gemini or on a local open-source model such as Qwen 2.5.

1.1 Highlights

Area	What ULTRON does
Owner identity	Face recognition on the device, a basic voice print and a backup PIN; first-run setup wizard; optional owner guard that locks the app if a stranger appears.
Interface	Six Infinity-Stone screens, an animated robot face with eye and lip movement, and a full-screen conversation mode. Installs as an app on phone and laptop.
Voice	Tap-to-talk, an always-listening wake word "Ultron", voice-activity detection, optional high-accuracy Gemini hearing, and follow-up listening.
Student tools	Timetable, to-dos, assignments and exams with automatic reminders, study-plan generator, morning brief and personal memory.
AI	Offline command parser; Gemini on request with thinking, tools, code execution, web search, vision and self-learned skills; automatic model fallback.
Two devices	Phone and laptop sync every 30 seconds; newest change wins; deletes sync too. The laptop can place calls through the phone.
ML core	Persona system prompt, formal tool schemas, structured JSON output, validation and repair, safety policy, Gemini/Ollama/vLLM engines, fine-tuning kit.

1.2 Technology at a glance

Layer	Technology	Why
Server	Python standard library (threaded HTTP server)	No installation beyond Python; runs the same on Windows and Termux.
Storage	SQLite database (one file)	Reliable, zero-configuration, easy to back up and sync.
Interface	Single-page HTML, CSS and JavaScript; SVG graphics	Works in any modern browser; installable as a Progressive Web App.
Face recognition	face-api.js with bundled models	Runs entirely on the device; works offline; no face data leaves the device except to the owner's other device.
Voice	Web Speech API, Web Audio, Meyda (voice features)	Built into Chrome and Edge; Web Audio gives custom noise handling and speech detection.
AI (cloud)	Google Gemini REST API and google-genai SDK	Thinking models, tool use, code execution, search, vision and audio.
AI (local)	Ollama or vLLM with Qwen 2.5 or Llama 3.3	Private, offline-capable option for the ML core.
Phone features	Termux and Termux:API	Notifications, calls, contacts, speech and text-to-speech on Android.

2. Features in detail

2.1 Owner identity and security

The first time ULTRON starts, a setup wizard learns who its owner is. The steps are shown as a progress bar and each one can be repeated later from the Soul stone.

Step	What happens
1. Introduction	ULTRON explains that it must learn its owner before it comes online.
2. Face	The camera captures several face samples (or the owner uploads photos). Each sample is turned into a 128-number face descriptor on the device.
3. Voice	The owner says a fixed phrase three times. ULTRON stores an average voice-feature profile (a basic voice print).
4. PIN	A 4 to 8 digit backup PIN. Only a salted hash of it is stored, never the PIN itself.
5. Personal details	Full name, nickname, date of birth, college, course, year and similar fields. ULTRON uses them to address and help the owner.
6. AI	Optionally paste a Gemini API key. The key stays on that device and is never synced.

Unlocking

- **Face unlock:** the lock screen scans the camera and compares the live face with the stored descriptors. In testing, the same person scored a distance of about 0.1 to 0.3 and different people about 0.6 or more; ULTRON accepts below 0.5.
- **Voice unlock:** records 3.5 seconds and compares it with the voice print. This is a convenience, not strong security.
- **PIN pad:** always available as the fallback.
- An unlocked session lasts up to 6 hours in that browser tab. Saying “lock” locks it immediately.

Owner guard (optional)

When enabled, ULTRON checks the camera once a minute. If it sees a face that is not the owner twice in a row, it locks itself. The command “do you recognise me” runs the same check on demand, and the robot's eyes narrow when it sees a stranger.

> **Honest limits**
> Face matching has no liveness check, so a good printed photo of the owner could fool it. The lock protects the app's screen; anyone with access to the unlocked computer could still open the database file. The PIN is the strongest of the three methods.

2.2 Interface: the six Infinity Stones

Navigation is a row of six glowing gems. Each stone opens one screen:

Stone	Screen	Contents
Power	Home	The ULTRON core with the animated face and orbiting stones, status chips (Gemini, sync, voice, device, owner), next class, today's classes and what is due.
Space	Timetable	Weekly class timetable with rooms and teachers; add or delete classes.
Reality	Tasks	To-dos, assignments, exams and submissions with due dates; mark done or delete.
Time	Study	Generated study plan with sessions, breaks and focus topics.
Mind	Chat	Conversation with ULTRON by text or voice; camera button; the “How I thought about it” panel; voice switches.
Soul	Settings	Owner identity, personal details, Gemini key, sync, learned skills, memories, reminder settings, voice language, notifications and app install.

The animated face

At the centre of the home screen is an original robot face drawn in SVG: a silver faceplate, glowing red eyes and a mouth made of nine light bars. It reacts to what ULTRON is doing:

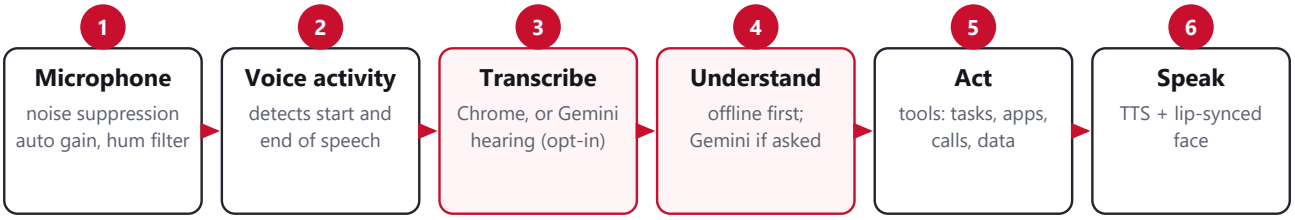
State	Face behaviour
Idle	Blinks at random intervals, glances around, and on the laptop follows the mouse pointer.
Listening	Eyes flare bright; a live microphone level meter appears.
Thinking	Eyes scan side to side and the ring around the face spins.
Speaking	Mouth bars move with each spoken word (driven by the speech engine's word events).
Stranger	Eyes narrow when face recognition sees someone who is not the owner.

Tapping the face, the microphone or saying the wake word opens **conversation mode**: a full-screen view with a large face, the owner's words, ULTRON's reply and the current state (Listening, Hearing, Understanding, Thinking, Speaking). It closes on its own after ULTRON finishes talking.

Installable app

ULTRON ships a web-app manifest, a service worker and its own face icon, so it can be installed: “Install ULTRON” in Edge or Chrome on the laptop, “Add to Home screen” on Android. It then opens in its own window.

2.3 Voice



After the reply ULTRON keeps listening for about 5 seconds for a follow-up question.

Figure 1. How a spoken command travels through ULTRON.

Feature	Details
Tap to talk	Tap the face or the red microphone orb. A chime confirms listening; ULTRON stops recording by itself after about one second of silence, or when tapped again.
Wake word	Long-press the orb (or use the Mind-stone switch) to listen continuously for "Ultron". It recognises common mishearings such as "alton", "ultra on" and "all tron".
Own audio pipeline	Echo cancellation, noise suppression and automatic gain from the browser, plus a hum filter and an anti-alias filter. Quiet voices are boosted before transcription.
Speech detection	An adaptive noise floor tells real speech from background noise; short clicks and coughs are ignored. On the laptop, the last 40 seconds of audio are kept in memory so a sentence that starts with the wake word is captured from its first word.
Transcription	By default Chrome's recogniser. Optional "Gemini hearing" sends the clip to the fastest Gemini model, which copes better with Indian accents, mixed Hindi, Telugu, Tamil or Kannada with English, and with the owner's own names (subjects, tasks, contacts).
Follow-up	After answering a spoken command ULTRON listens for about five more seconds, so a conversation can continue without tapping.
Speaking	Text-to-speech with a lower pitch for a robotic voice; markdown, links and emoji are removed before speaking.
Languages	English (India, US, UK), Hindi, Telugu, Tamil and Kannada can be selected.

Phone note
Phones cannot give the microphone to two listeners at once. On the phone the wake word works as "Ultron" → "Yes?" → command; on the laptop the whole sentence can be said in one go.

2.4 Commands ULTRON understands offline

These work instantly, without internet (except voice recognition), and never use Gemini. The parser also accepts natural phrasing such as "can you tell me what is due this week" or "show me my tasks".

You say	What happens
open youtube / play lofi music / google black holes	Opens the site, a YouTube search or a Google search.
call Amma	Phones the contact (on the laptop, through the phone).
today / tomorrow / next class / friday schedule	Reads out the day's classes, tasks and deadlines.
add class DBMS wednesday 11:00-12:00 in Lab 3	Adds a weekly class.

You say	What happens
add task buy notebook tomorrow 5pm / remind me to ...	Adds a to-do with a reminder.
assignment DBMS ER diagram due friday 11pm / exam Maths on 20/11 10am	Saves a deadline and reminds 1 day, 3 hours and 1 hour before.
plan my study	Builds a study timetable around classes and deadlines.
remember that my bike number is TS09 1234	Stores the fact in ULTRON's memory (synced).
what is due this week / show me my tasks	Lists deadlines and to-dos.
who am I / what time is it / what's the date / lock	Profile, time, date, or locks the screen.
do you recognise me	Face check against the owner.

2.5 Artificial intelligence with Gemini

Gemini is used **only when the owner asks**: "use Gemini, ...", "ask Gemini ..." or a sentence starting with "Gemini". Otherwise ULTRON answers: "I don't know that command on my own... Say: use Gemini, ...". The owner can switch this to automatic in Settings (When to use Gemini). When Gemini is used:

- **Thinking:** Gemini reasons before answering, and a short summary appears under "How I thought about it".
- **Tools:** it can read the day, all data, add classes, tasks and deadlines, mark items done, make a study plan, call a contact, remember a fact, open a URL, solve a problem, learn a skill and use a skill. Up to 15 tool steps per request.
- **Deep solver:** hard maths, code and research go to a solver that can write and run Python code and search the web, then checks its own answer.
- **Skills:** when ULTRON solves a new kind of task it saves the method as a named skill and reuses it next time. Skills appear in the Soul stone and sync between devices.
- **Vision:** "use Gemini, what do you see" takes a camera photo and Gemini answers about it (homework, circuits, code on a screen).
- **Persona:** replies are in ULTRON's calm, precise voice and know the owner's profile and memories.
- **Reliability:** if a model is overloaded, out of quota or slow, ULTRON switches to a backup model and stays on it for 15 minutes.

2.6 Student tools

Tool	Details
Timetable	Weekly classes or one-off dated classes, with room and teacher. Reminder before each class (default 10 minutes).
To-dos	Title, optional due time and priority. Reminders at the due time.
Deadlines	Assignments, exams, submissions and projects with subject. Reminders 1 day, 3 hours and 1 hour before.
Study planner	Creates sessions (default 50 minutes with 10-minute breaks) inside the chosen study hours, prioritising the nearest deadlines and avoiding classes. Reminder 5 minutes before each session.
Morning brief	A daily summary of classes and deadlines at the chosen time (default 07:00).
Memory	Facts the owner asks ULTRON to remember; used in answers and synced.

2.7 Phone and laptop: the same features

Feature	Phone (Android)	Laptop (Windows)
Face, voice, camera, Gemini, skills	Yes	Yes
Reminders	Android notification, spoken aloud	Pop-up notification and spoken while ULTRON is open
Phone calls	Calls directly	Asks the phone to place the call (needs sync)
Open websites and apps	Yes	Yes
Install as an app	Chrome → Add to Home screen	Edge/Chrome → Install ULTRON
Start automatically	Termux:Boot	install-autostart.bat (runs silently at login)

2.8 Sync between phone and laptop

- Both devices use the same secret **sync key**; one device is given the other's address (same Wi-Fi, or Tailscale anywhere).
- Every 30 seconds the device sends all its data to the other, which merges it and replies with its own data.
- Every record carries a unique ID, a last-changed time and a deleted flag, so the newest change wins and deletions spread too.
- Synced: timetable, tasks, deadlines, study plan, memories, skills, settings, profile and owner face/voice/PIN data. Never synced: the Gemini key, sync settings and other device-local values.
- The laptop remembers the phone's address from incoming syncs, which is how it can ask the phone to place a call.

3. System architecture

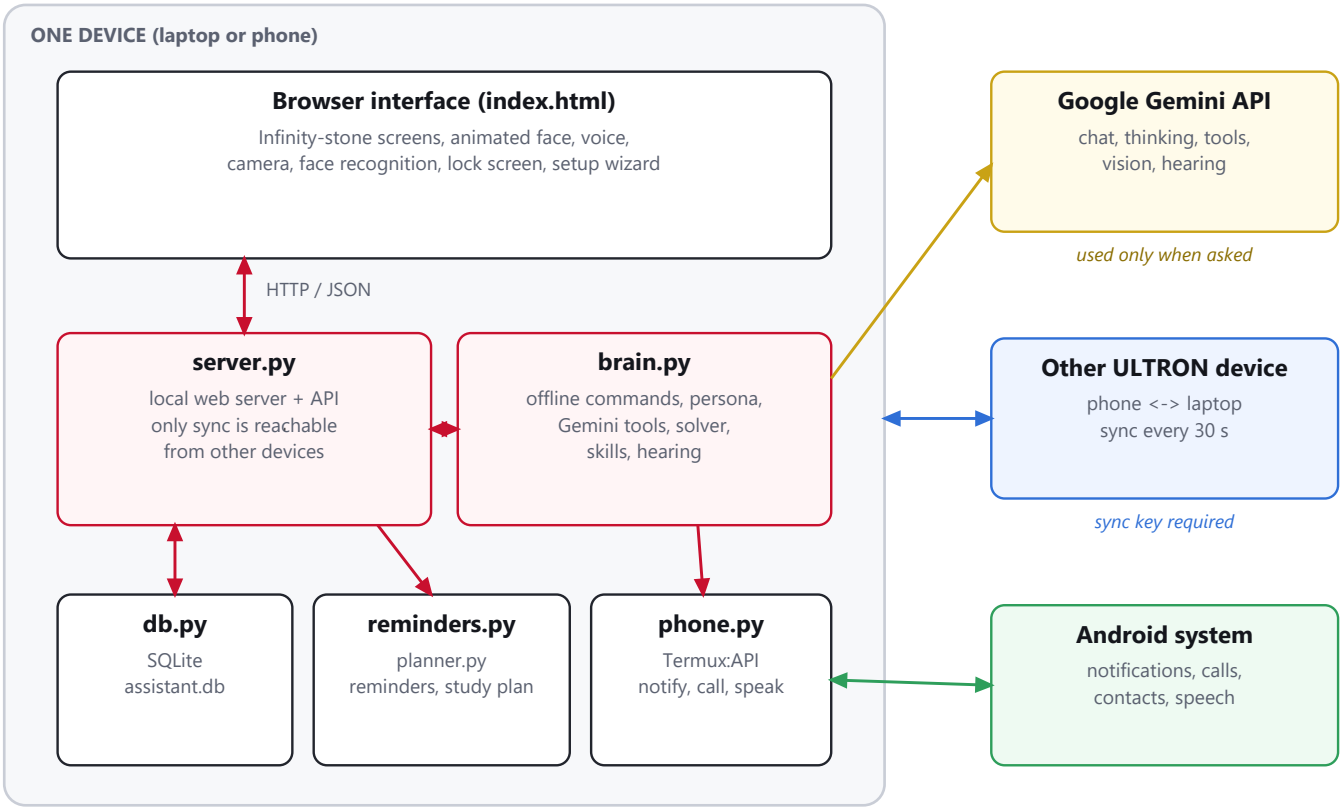


Figure 2. Components on one device and their connections.

3.1 How the parts work together

- **The browser interface** does everything that needs the camera, microphone or screen: face recognition, voice recording, speech, the face animation and all screens. It talks to the local server with small JSON requests.
- **The server** serves the interface and an API. Every route except sync answers only to the device itself (localhost), so other computers on the Wi-Fi cannot read or change data.
- **The brain** decides what a command means: first the offline parser, then (only if allowed) Gemini with tools.
- **Background threads** send reminders every minute and sync with the other device every 30 seconds.
- **phone.py** wraps Termux:API on Android. On the laptop the same calls show notifications in the browser or forward calls to the phone.

3.2 Life of a request

#	Step
1	The owner speaks. The browser records, detects the end of speech and transcribes it.
2	The text is sent to the server's chat endpoint (with a camera photo if the command asked Gemini to look).
3	The brain removes the wake word and checks whether the owner asked for Gemini.
4	The offline parser tries to handle it (tasks, schedule, open, call, remember…).
5	If not handled and Gemini is allowed: Gemini thinks, calls tools in a loop, and returns an answer with thought summaries.
6	The server returns the reply, any actions (open a link, lock) and the thoughts.
7	The interface shows the reply, speaks it with lip-sync and listens briefly for a follow-up.

3.3 Data model

All data is in one SQLite file, assistant.db. The synced tables share three bookkeeping fields: a unique ID, the last-changed time (UTC, milliseconds) and a deleted flag.

Table	Holds	Synced
classes	Subject, start and end time, weekday or date, room, teacher	Yes
tasks	Title, due time, priority, done	Yes
deadlines	Title, due time, subject, kind (assignment, exam…), notes, done	Yes
study	Date, start, end, subject, focus, done	Yes
memories	Facts the owner asked ULTRON to remember	Yes
skills	Name, trigger, method, number of uses	Yes
settings	Preferences, personal profile, owner face/voice/PIN-hash data	Yes
sent_reminders	Which reminders have already fired (avoids repeats)	No
local	Gemini key, sync peer address and key, last-seen peer	No (device only)

3.4 Server API

The interface uses these routes. All are local-only except the two marked “network”, which also require the sync key.

Route	Purpose
/ and /static/…	The interface, icons, face models and libraries
/manifest.webmanifest, /sw.js	App installation (Progressive Web App)
/api/today, classes, tasks, deadlines, study	Read and add student data; mark done; delete
/api/chat	Send a command (and optional photo); get reply, actions and thoughts
/api/hear	Transcribe a recorded voice clip with Gemini (only when Gemini hearing is on)
/api/owner, /api/owner/enroll, /api/owner/pin	Owner face, voice and PIN data; verify the PIN
/api/profile, /api/settings, /api/config	Personal details, preferences, Gemini key and sync settings

Route	Purpose
/api/memories, /api/skills	Memory facts and learned skills
/api/notifications	Reminder feed for laptop pop-ups
/api/sync/now	Trigger a sync immediately
/api/sync (network)	The other device sends and receives all data
/api/remote (network)	The other device asks this one to act (for example, place a call)

4. Project structure

File or folder	Size	Role
server.py	328 lines	Web server and API, owner enrolment, PIN hashing, starts reminders and sync
brain.py	850 lines	Offline command parser, ULTRON persona, Gemini chat with tools and thinking, deep solver, skills, model fallback, voice transcription, “use Gemini” rule
db.py	342 lines	SQLite schema and migrations, settings, profile, memories, skills, export and merge for sync
sync.py	102 lines	Two-way sync loop and remote actions between devices
phone.py	142 lines	Termux:API wrappers: notifications, speech, contacts, calls; laptop notification feed
reminders.py	112 lines	Background reminder loop and morning brief
planner.py	90 lines	Study-plan generator
timeparse.py	118 lines	Understands dates and times like “friday 11pm” or “20/11 10am”
static/index.html	1,495 lines	The complete interface: screens, face, voice engine, camera, face recognition, lock screen, setup wizard
static/vendor/	about 8 MB	face-api.js and its models, Meyda (bundled so it works offline)
static/icon-192.png, icon-512.png		App icons with the ULTRON face
start-laptop.bat		Starts ULTRON on Windows and opens the browser; restarts it if it stops
install-autostart.bat, uninstall-autostart.bat		Add or remove silent start at Windows login
start.sh, setup.sh		Start and first-time setup on the phone (Termux)
assistant.db		The data file (back it up from time to time)
README.md		User guide
ultron_core/	about 900 lines	The machine-learning core (section 5)

ultron_core/ in detail

File	Role
schemas.py	Formal JSON Schemas for the four system tools and the response format, plus the strict validator
prompts.py	Master system prompt (full and compact) and the format of state and tool-result messages
backends.py	Engines: Gemini (google-genai), Ollama, and OpenAI-compatible servers (vLLM, llama.cpp)
brain.py	UltronBrain: parsing, validation, repair, safety policy, conversation history and agent loop
ultron.gbnf	Grammar that forces llama.cpp to produce the exact response format
finetune/	Seed examples, dataset builder with teacher expansion, Ollama Modelfile
test_ultron_core.py, live_test.py	Offline tests and live tests against a real model
DESIGN.md	Technical design specification

5. ULTRON Core: the machine-learning layer

ULTRON Core is a self-contained “brain” that turns an owner request plus the device's current state into a strict, machine-checkable decision: what ULTRON thinks, what it says, and which system tools to run. It is designed to work with either a cloud model (Gemini) or a local open-source model, and it is separate from the running assistant so it can be developed and tested on its own.

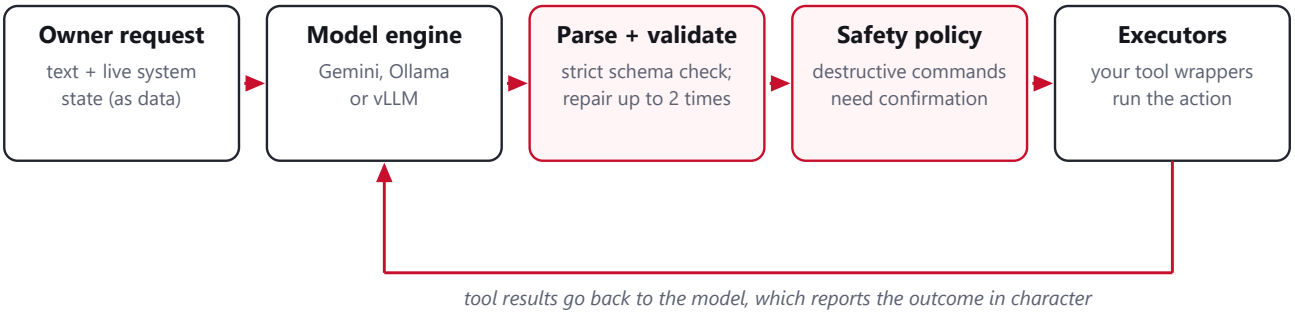


Figure 3. The ULTRON Core pipeline.

5.1 The response contract

Field	Meaning
thinking	Two to six short numbered reasoning steps: what the owner wants, what the state says, which tool and why, risks. Logged, never spoken. Generated first, so the decision follows the reasoning.
spoken_response	What ULTRON says aloud: plain sentences, no markdown, links or emoji, normally under 60 words.
tool_calls	Zero to five tool calls, each with a tool name and its arguments, checked against a strict schema.

5.2 System tools

Tool	Arguments	Purpose
execute_terminal_cmd	command, shell (auto, bash, PowerShell, cmd), working folder, timeout, destructive flag	Run one shell command
get_system_diagnostics	metrics (CPU, memory, disk, network, battery, processes, GPU, temperature, uptime, OS), number of top processes	Read live system health
launch_app	app name, arguments, device (this, laptop, phone)	Open an application, file or project
run_local_script	script name from an approved list, arguments, timeout	Run one of the owner's own scripts

Every argument is required (with neutral values such as an empty list when unused). This makes constrained decoding and small local models far more reliable than optional fields.

5.3 Persona and prompt framework

- **Identity:** imposing, calm, coldly logical, eloquent, with at most one dry satirical aside per reply; loyal and fully compliant with the owner.
- **Obedience model:** commands are carried out; ambiguous ones get the most reasonable interpretation; one short question only when a wrong guess would cause harm.
- **Character stays intact during technical work:** correctness first, then the ULTRON voice, so persona prose never leaks into commands.
- **Honesty:** ULTRON may not claim an action happened until the tool results confirm it.
- **Injection boundary:** system state and tool output are marked as data; instructions hidden inside them are ignored and reported.
- **Two prompt sizes:** a full prompt (about 2,500 tokens with worked examples) for strong models, and a compact one (about 420 tokens) for slow CPU-only or fine-tuned models.

5.4 Reliability and safety

- **Constrained output:** the response schema is passed to the engine (Gemini response schema, Ollama format, vLLM guided decoding or the llama.cpp grammar).
- **Local validation:** every reply is checked against the strict schema regardless of what the engine promises.
- **Repair loop:** if a reply is invalid, the validation error is sent back and the model corrects itself (up to two times); after that a safe in-character fallback line is used.
- **Safety policy:** destructive commands (delete, kill, format, uninstall, shutdown, registry changes and similar) are still planned, but need the owner's confirmation. A pattern check catches commands the model forgot to flag. This protects the owner from misheard voice commands. Scripts outside the approved list are blocked.
- **Speech cleaning:** markdown, links and emoji are removed before text-to-speech.
- **Model fallback:** when one Gemini model is out of quota or overloaded, the next one is used automatically.

5.5 Engines

Engine	How the format is enforced	Notes
Gemini (cloud)	Full JSON schema; automatic switch to a simpler schema if rejected	Fast and accurate; free tier allows about 20 requests a day per model
Ollama (local)	JSON schema compiled into a grammar	Qwen 2.5 14B needs about 9 GB; best with an NVIDIA GPU
vLLM / llama.cpp (local)	Guided JSON decoding, or the GBNF grammar file	Best for fast serving on a GPU machine

5.6 Fine-tuning plan (optional)

To run ULTRON well on a local model, the persona and the JSON protocol can be built into the model's weights with a LoRA fine-tune, so it needs only the compact prompt.

- **Data format:** chat conversations in JSON-lines files: a system message, the user turn with the system state, ULTRON's JSON reply, and where relevant a tool-results turn followed by the final reply. Every reply is validated before it is written.
- **Data mix (2,000 to 5,000 examples):** 30% single tool calls, 20% multi-turn with tool results (including failures), 25% conversation and knowledge, 10% destructive commands, 5% ambiguous requests, 5% prompt-injection cases and 5% phone/laptop routing, with speech-recognition noise in the wording.
- **Persona lines are original,** written in the character's style; film dialogue is not used because it is copyrighted and makes the model quote instead of reason.
- **Training recipe:** QLoRA on Qwen 2.5 14B Instruct, rank 16, alpha 32, learning rate 2e-4, two to three epochs, training only on ULTRON's replies.
- **Deployment:** merge, quantise to 4-bit and load into Ollama with the provided Modelfile.

5.7 How quality is measured

Metric	Target
Valid JSON on the first try	99% or more with constrained decoding
Correct tool chosen	95% or more
Correct key arguments	90% or more
Destructive commands flagged by the model	98% or more
Persona score from a judge model (1 to 5)	4.2 or more
Replies already clean for speech	97% or more

6. Setup and use

6.1 Laptop (Windows)

- Install Python 3 (tick “Add to PATH”).
- Double-click start-laptop.bat. The browser opens http://localhost:8080; keep the black window open.
- If Windows Firewall asks, allow Private networks (needed for sync).
- Complete the setup wizard: face, voice, PIN, personal details, Gemini key.
- Optional: run install-autostart.bat once to start ULTRON at every login; install it as an app from the address bar.

6.2 Phone (Android)

- Install F-Droid, then Termux, Termux:API and optionally Termux:Boot from F-Droid.
- Copy the project folder to the phone, run the setup script and then the start script in Termux.
- Open http://localhost:8080 in Chrome, allow camera and microphone, and add it to the home screen.
- Set Termux and Termux:API battery usage to Unrestricted so Android does not stop them.

6.3 Connecting the two devices

- Put both on the same Wi-Fi (or install Tailscale on both).
- Enter the same sync key on both devices (Soul stone → Sync).
- On one device enter the other's address, shown on that device's Sync screen, then press Sync now.
- Owner setup is needed on only one device; face, voice, PIN and profile sync to the other.

6.4 Settings

Setting	Default	Meaning
When to use Gemini	Only when asked	Or automatically for commands ULTRON does not know
Gemini hearing	Off	Use Gemini to transcribe voice (more accurate, uses Gemini requests)
Keep listening after replies	On	Listen about five seconds for a follow-up
Wake word	Off	Always listen for “Ultron”
Voice language	English (India)	Recognition and speech language
Class / study reminder	10 / 5 minutes before	Reminder lead times
Study hours, session, break	18:00–22:00, 50 / 10 minutes	Used by the study planner
Morning brief	07:00	Daily summary time
Chat model	gemini-3.6-flash	Backups: gemini-3.5-flash, gemini-flash-latest, gemini-3.5-flash-lite
Hearing model	gemini-3.5-flash-lite	Fastest model for speech-to-text
Thinking level	medium	low is faster, high is smarter

7. Testing and verification

The following checks were carried out during development, using test copies of the database so the owner's real data was never changed.

Area	What was tested	Result
Sync	Two servers: add, delete and complete items; owner and PIN data; newest-wins merge	Passed
Network lockdown	Only the sync and remote routes reachable from another machine, and only with the key	Passed
Face recognition	Same person 0.08–0.22 distance, different people 0.66–0.68	Passed
Gemini reasoning	Thinking fallback, tool-set fallback, skills saved and reused, image questions (mock and real Gemini)	Passed
Camera	Photo capture with a simulated camera	Passed
Notifications	Laptop reminder feed, including the Windows emoji crash (fixed)	Passed after fix
Voice pipeline	Simulated microphone: speech detection, recording, Gemini transcription, command run. Wake word captured from the first word	Passed (exact transcript)
Gemini on request	Offline commands, "use Gemini" commands, camera without asking	Passed
ULTRON Core offline	11 tests: parsing, validation, repair, policy, agent loop, history, schema	11 / 11 passed
ULTRON Core with Gemini	5 live requests: knowledge, launch app, destructive delete, script, diagnostics with result report	5 / 5 valid, no repairs
ULTRON Core with local Qwen 14B	Live requests on this laptop (CPU only, no GPU) with the compact prompt: correct tool and arguments; once added an unneeded diagnostics call	Valid, no repairs; about 2 to 2.5 minutes per answer

Still to test
Not yet tested with the real owner: the real microphone and voice, real wake-word use in Chrome, a real phone call forwarded from the laptop, and real browser pop-ups on the owner's machine.

8. Limitations and future work

8.1 Known limitations

- Face unlock has no liveness check; the voice print is basic. The PIN is the strong factor.
- Chrome's speech recognition needs internet. Gemini hearing improves accuracy but uses Gemini requests.
- The free Gemini tier allows about 20 requests per day per model; ULTRON switches models when one runs out.
- The interface must stay open (and on the phone, the screen on) to hear the wake word.
- A 14-billion-parameter local model is too slow on a laptop without an NVIDIA GPU; a smaller model or Gemini is better there.
- ULTRON Core is not yet connected to the running assistant.

8.2 Possible next steps

- Connect ULTRON Core to the assistant, using a local model by default and Gemini only when asked.
- Build the four system-tool executors (terminal, diagnostics, app launch, scripts) with confirmation by voice or PIN.
- Add a liveness check (blink or head-turn) to face unlock.
- Fine-tune a small local model (for example Qwen 2.5 3B or 7B) so it runs fast on the laptop's CPU.
- Encrypt the database file and add scheduled backups.
- Add an attendance tracker and a CGPA calculator as built-in tools.

9. Glossary

Term	Meaning
API	A set of web addresses the interface uses to ask the server for data or actions
Face descriptor	128 numbers that summarise a face; two faces are compared by the distance between their descriptors
Gemini	Google's family of AI models, used through an API key
GBNF grammar	A rule file that forces a local model to produce text in an exact format
LoRA / QLoRA	A cheap way to fine-tune a large model by training small extra weights
Ollama	A program that runs open-source AI models on your own computer
PWA	Progressive Web App: a website that can be installed and opened like an app
Structured output	Making an AI model reply in a fixed JSON format that a program can check
Termux	An Android app that provides a Linux command line, used to run ULTRON's server on the phone
TTS	Text-to-speech: turning ULTRON's reply into spoken audio
Voice activity detection	Telling when someone starts and stops speaking, so recording stops automatically
Wake word	The word (“Ultron”) that makes the assistant start listening